Language-conditioned Spatial Affordance Refinement for Diffusion-based Grasp Detection

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https://github.com/TBLboy/language-driven-grasp-detection

Abstract

Language-driven grasp detection requires grounding a natural-language grasping instruction to a manipulable region in an RGB image. We build on the official LGDM baseline and repair two implementation gaps: the ALBEF text condition is not connected to the grasp decoder, and the diffusion loss is not back-propagated. We then propose LSAR, a lightweight spatial affordance refinement module that injects a small residual correction into the decoder bottleneck before grasp generation. On a scene-disjoint 10,000-scene subset of Grasp-Anything++, LSAR improves repeated-evaluation correct grasps from 470.0 to 678.0 and 661.7 under two training seeds, while preserving the official grasp representation and evaluation protocol.

1. Introduction

Language-driven grasp detection requires predicting a five-parameter grasp rectangle

$$G = (x, y, w, h, \theta) \tag{1}$$

for the object or part selected by a free-form instruction. We start from the diffusion-based LGDM formulation. In its released code, the vision-language fusion output is computed but not injected into the grasp decoder, and the diffusion loss is not optimized. We define a clean baseline that restores the diffusion objective and introduce LSAR, a compact residual module that makes the language condition spatially explicit. The main contribution is the LSAR conditioning mechanism; the repaired baseline, repeated evaluations, and ablation are provided to make the effect reproducible and interpretable.

2. Related Work

Grasp detection methods based on dense quality, angle, and width maps [1, 3] provide the representation used by many language-conditioned systems. Language-driven Grasp Detection (LGD) [4] introduces Grasp-Anything++ [5] and a diffusion-based conditional model. Vision-language encoders such as ALBEF [2] align semantic concepts but do not directly encode graspable regions. Our module adapts the language-conditioned spatial branch without changing the pretrained encoders or the generative head.

3. Method

3.1. Problem Setup

An input sample is (I, T), where I is an RGB image and T is a grasping instruction. Ground-truth rectangles are converted into dense maps pos, $\cos(2\theta)$, $\sin(2\theta)$, and normalized width; post-processing decodes the predicted dense maps into a rectangle.

3.2. Clean LGDM Baseline

LGDM computes $y_{\text{view}} \in \mathbb{R}^{8\times 19\times 19}$ with ALBEF, but the released decoder does not use it, and the upstream training script does not back-propagate the diffusion loss. Our baseline keeps the official decoder structure and optimizes

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{diffusion}} + \mathcal{L}_{\cos} + \mathcal{L}_{\sin} + \mathcal{L}_{\text{width}}. \tag{2}$$

We also re-enable a raw "plain-y" injection into the GG-CNN decoder feature F_v for the language-conditioning ablation.

3.3. LSAR

Semantic alignment does not identify which part of a target affords a grasp. LSAR combines the decoder feature with the language map and learns a small residual:

$$Z = \operatorname{concat}\big(F_v,\ y_{\text{view}}\big), \tag{3}$$

$$H = \operatorname{ReLU}\big(\operatorname{conv}_{3\times3}(\operatorname{GN}(\operatorname{ReLU}(\operatorname{conv}_{3\times3}(Z))))\big), \tag{4}$$

$$\Delta = s\,\operatorname{conv}_{1\times1}(H), \tag{5}$$

$$F' = \operatorname{ReLU}\big(F_v + \Delta\big). \tag{6}$$

We fix $s = 0.01$ to keep the residual perturbation small. The module also emits a 19×19 affordance map $A = \operatorname{conv}_{1\times1}(H)$ for an auxiliary term studied below.

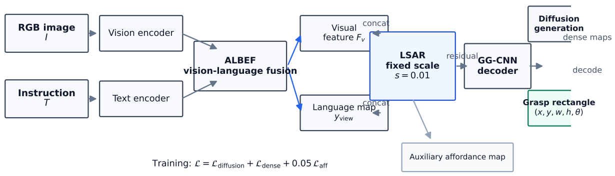

Figure 1. Pipeline: ALBEF fusion, LSAR residual injection, GG-CNN decoding, and diffusion grasp generation.

Table 1. Repeated 10-step evaluation on 2,000 validation stems. Final LSAR uses scale 0.01 and $\lambda_{\text{aff}} = 0.05$.

Method	λ_{aff}	Correct	Mean $\pm$ Std
LGDM baseline	—	449	470.0 ± 10.8
LSAR-no-aff	0.0	643	653.7 ± 6.5
LSAR-full	0.1	625	605.3 ± 18.9
LSAR (seed 42)	0.05	686	$\mathbf{678.0} \pm 16.5$
LSAR (seed 43)	0.05	666	661.7 ± 14.0

3.4. Training Objective

The final objective is

$$\mathcal{L} = \mathcal{L}_{\text{base}} + \lambda_{\text{aff}}\,\text{MSE}\Big(A,\ \text{AdaptiveAvgPool}_{19}(\text{pos})\Big), \tag{7}$$

with $\lambda_{\text{aff}} = 0.05$ in the final method.

4. Experiments

4.1. Data and Protocol

We use 10,000 Grasp-Anything++ stems with one unique scene per stem and split 8,000 train / 2,000 validation with seed 42, so the split is scene-disjoint by construction. All models train for 15 epochs, batch size 2, learning rate 10^{-3}, weight decay 10^{-4}, and evaluation uses 10-step respaced diffusion sampling. A prediction is correct if it matches any ground-truth rectangle with IoU > 0.25 and angle error $< 30°$.

4.2. Results

Table 1 reports one training-loop evaluation and the mean over three sampling evaluations. The final method improves the baseline mean from 470.0 to 678.0 (seed 42) and 661.7 (seed 43). Removing the auxiliary affordance loss gives 653.7; the larger weight $\lambda = 0.1$ gives 605.3. LSAR therefore contributes mainly through spatial conditioning, while the small affordance term is a useful auxiliary objective. On a fixed 200-sample subset, 50 instead of 10 diffusion steps improves the baseline from 39 to 44 correct and final LSAR from 65 to 67, preserving the ordering.

Figure 2 shows paired part-level examples. The conditioned model correctly locates a highlighter and keychain keys where the baseline fails and retains success on an apple-stem case. This illustrates that the refinement improves spatial grounding rather than only global image semantics.

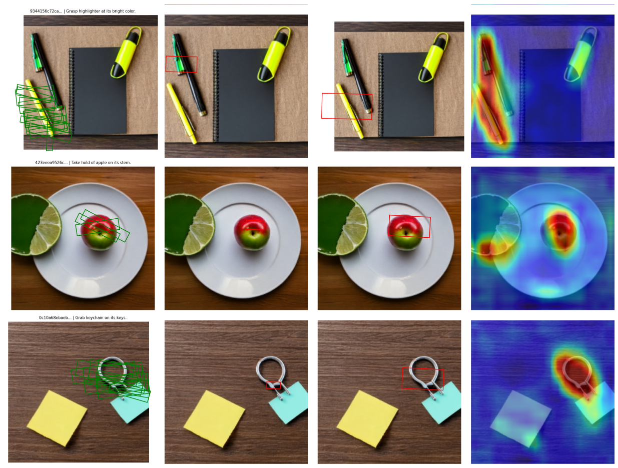

Figure 2. Paired qualitative results: input, ground truth, LGDM prediction, LSAR prediction, and LSAR affordance map.

5. Conclusion

We present a compact and reproducible language-driven grasp detection system based on LGDM and LSAR. The LSAR residual conditioning improves spatial grounding and consistently increases correct-grasp rate over the repaired baseline on a 10,000-scene Grasp-Anything++ subset. Results are limited to a subset, 15 training epochs, and 10-step sampling; full-scale training and real-robot validation are left as future work.

References

[1] Sulabh Kumra, Shirin Joshi, and Ferat Sahin. Antipodal robotic grasping using generative residual convolutional neural network. In Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020.

[2] Junnan Li, Ramprasaath Selvaraju, Akhilesh Gotmare, Shafiq Joty, Caiming Xiong, and Steven Hoi. Align before fuse: Vision and language representation learning with momentum distillation. In Advances in Neural Information Processing Systems (NeurIPS), 2021.

[3] Douglas Morrison, Peter Corke, and Jürgen Leitner. Closing the loop for robotic grasping: A real-time, generative grasp synthesis approach. In Robotics: Science and Systems (RSS), 2018.

[4] An Dinh Vuong, Minh Nhat Vu, Baoru Huang, Nghia Nguyen, Hieu Le, Thieu Vo, and Anh Nguyen. Language-driven grasp detection. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

[5] An Dinh Vuong, Minh Nhat Vu, Hieu Le, Baoru Huang, Binh Huynh, Thieu Vo, Andreas Kugi, and Anh Nguyen. Grasp-anything: Large-scale grasp dataset from foundation models. In Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2024.